

Encryption

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Today's topics

Hey, check out these slides. . .

- What is *encryption*
- What is a *cryptosystem*
- Types of encryption
 - Symmetric encryption
 - Asymmetric encryption
- Public and Private keys
- Key Exchanges

What is Encryption

Encryption is the process of *encoding* information with a purpose (typically for security security). This process converts the original representation of the information, known as *plaintext*, into an alternative form known as *ciphertext*.

This process, combined with its reverse *decryption*, and all of the data/algorithms necessary to encrypt and decrypt messages form a *cryptosystem*.

-Wikipedia

Encoding and Encryption



Figure 1: Scytale

What is Encryption (continued)

Some points:

- Encryption is reversible (with a key)
- Encryption is everywhere
- Encryption does NOT need a computer (but good encryption probably does. . .)

Cryptosystem

Five-tuple (sequence) of the following elements:

Element	description
E	the set of Encryption algorithms
D	the set of Decryption algorithms
M	the set of plaintext Messages
K	the set of Keys
C	the set of encrypted messages or Ciphertexts

$$E = f(K, M) \rightarrow C$$

$$D = f(K, C) \rightarrow M$$

Scytale Cryptosystem

Element	description
<i>E</i>	wrap around scytale & write <i>message</i>
<i>D</i>	wrap around scytale & read <i>message</i>
<i>M</i>	the <i>message</i>
<i>K</i>	???
<i>C</i>	strip or paper with letters

Caesar Cipher

AKA rotational cipher, shift cipher, probably more. Simple encryption via substitution.

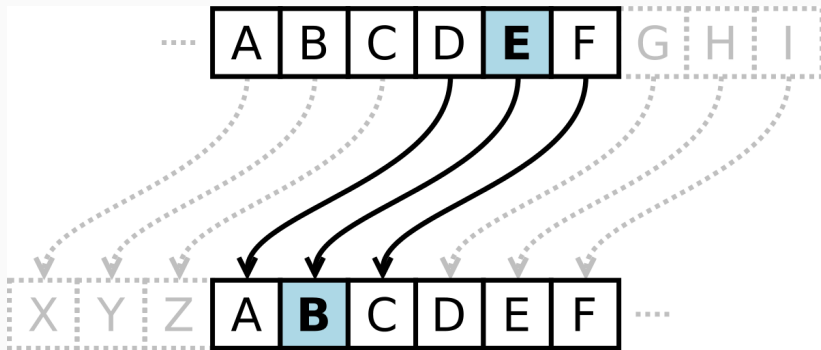


Figure 2: caesar cipher

Lets get Math-y

Element	description
<i>M</i>	<i>? ?? ????</i>
<i>C</i>	<i>F XJ EBOB</i>
<i>E</i>	$C[n] = (M[n] - 3) \text{ Mod } 26$
<i>D</i>	$M[n] = (C[n] + 3) \text{ Mod } 26$
<i>K</i>	What is the key?



$M[n] = \text{ABCDEFGHIJKLMNOPQRSTUVWXYZ}$

$C[n] = \text{XYZABCDEFGHIJKLMNOPQRSTUVWXYZ}$

Two types of Encryption

Symmetric

Same key for both encrypting and decrypting

Asymmetric

Different keys for encrypting and decrypting

Symmetric Encryption

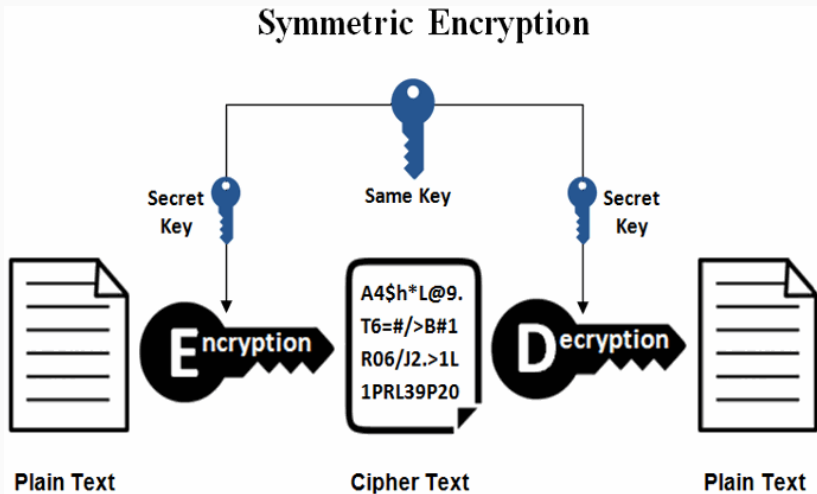


Figure 3: symmetric key encryption

Keys are identical for encryption and decryption.

Asymmetric Encryption

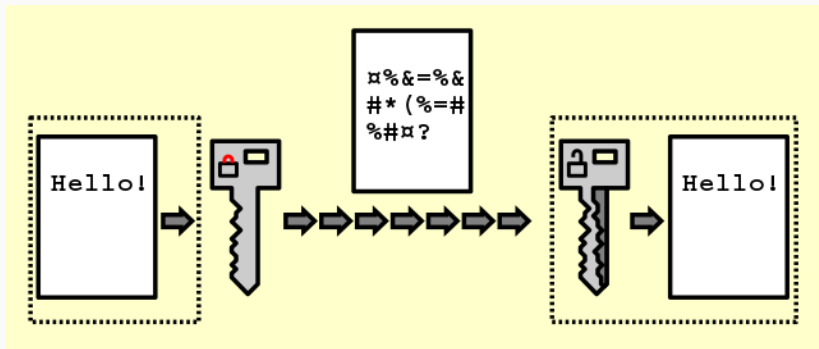


Figure 4: public key encryption

Keys are different but mathematically related.

Differences

Symmetric	Asymmetric
* One key * Fast * Used to encrypt various amounts of data * Key must be secured * Needs more keys to establish secure communications within a group	* Multiple keys * Slow * Used to authenticate and initiate symmetric encryption * only private key needs securing * Needs fewer keys to establish secure communication in a group

Apples and Oranges

Although the end goal is the same (end up with *encrypted* data in the form of a cipher text), asymmetric and symmetric encryption work very differently.

	Asymmetric (RSA 2048)	Symmetric (AES 128)
<i>Encrypt</i> spd	1.5 MB/s	100 MB/s
<i>Decrypt</i> spd	0.4 MB/s	100 MB/s

Public and Private Key pairs

- Public key is public information, can be shared with anyone.
- Assume **everyone** has your public key
- Private key must be kept private
- **Public** and **Private** key pairs share a unique mathematical relation:
 - Anything **encrypted** with the **public key** can only be *decrypted* with the *private key*
 - Anything **encrypted** with the **private key** can only be *decrypted* with the *public key*

Key Pairs continued

If we trust this relationship, and we trust that recorded knowledge of who has what keys:

- We can check *Authenticity* by asking them to encrypt something with their private key
- We can receive (one way) secure communications (ask someone else to first encrypt with your public key)
- If we send something encrypted with our private key who can read it???

Diffie-Hellman Key Exchange

Solves the problem of securely exchanging cryptographic keys over a public channel.

- The internet is public and filled with many potential eavsdroppers (***Eve's***)



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- How to establish a *secure* communication channel across an insecure one?

Diffie-Hellmen

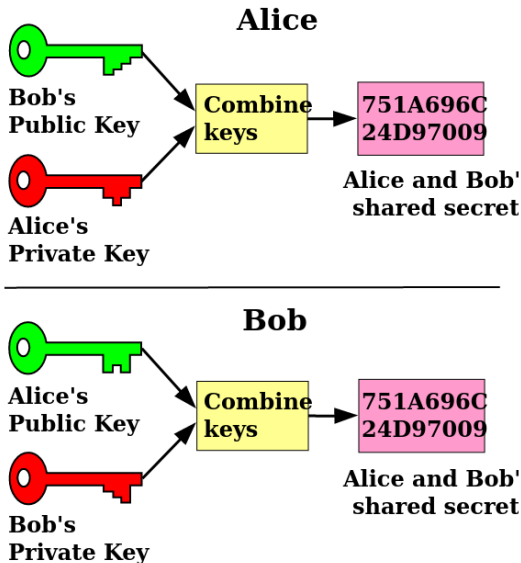


Figure 5: Diffie-Hellman

Diffie-Hellman w/colors

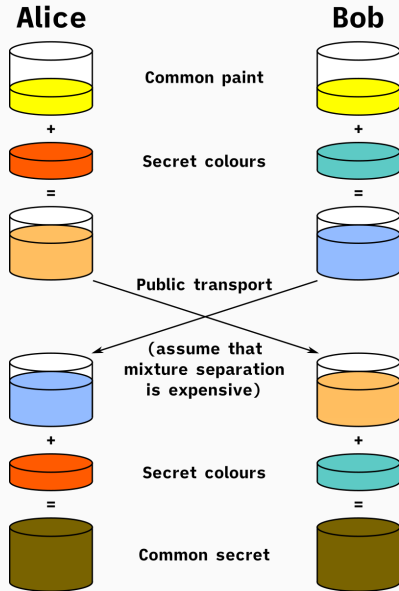


Figure 6: Diffie-Hellman with colors

Diffie-Hellman usage

HTTPS://

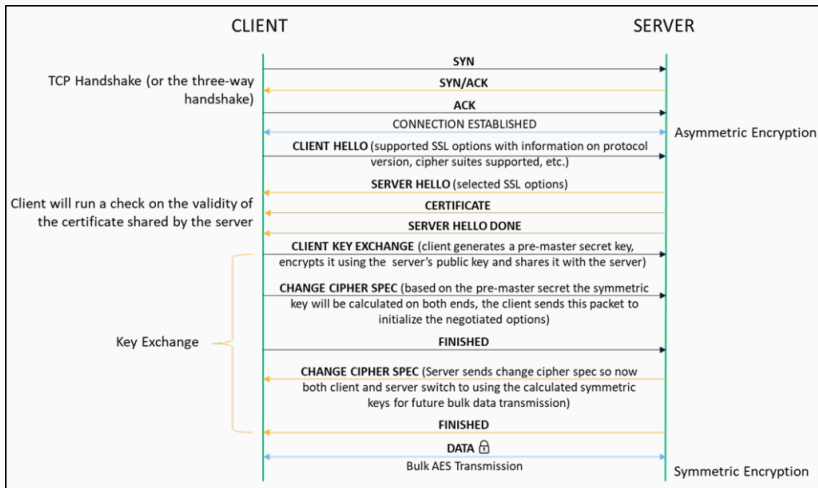


Figure 7: TCP Handshake

